Erwin Quanten

Full-Stack AI Engineer – React Python

Brussels, Belgium | erwin.quanten.dev@outlook.com | +32460256356 | https://www.linkedin.com/in/erwin-quanten-b82815a/

PROFESSIONAL SUMMARY

Staff Software Engineer with 16+ years delivering enterprise **AI, SaaS, cloud, and investment workflow platforms**, specializing in **React 18.x**, **Next.js 13.x–14.x**, **Python 3.10–3.12**, **FastAPI 0.100.x–0.110.x**, **Azure OpenAI**, **RAG**, **agentic workflows**, and production API architecture. Experienced building secure full-stack systems for private-market due diligence, portfolio analytics, deal execution, and risk monitoring, improving retrieval quality, release reliability, and stakeholder productivity across Azure-based enterprise environments while modernizing legacy services and aligning AI features with investment operations teams securely.

SKILLS

- **Frontend: React 18.x**, **Next.js 13.x–14.x**, **TypeScript 4.9–5.x**, Redux Toolkit, React Query, Zustand, Tailwind CSS, Material UI, WebSockets
- **Backend: Python 3.10–3.12**, **FastAPI 0.100.x–0.110.x**, Flask 2.x–3.x, Django 4.x–5.x, Node.js 18.x–20.x, REST APIs, GraphQL, OpenAPI
- **AI / LLM: Azure OpenAI**, OpenAI APIs, **RAG pipelines**, LangChain 0.1.x–0.2.x, LangGraph 0.1.x, prompt engineering, evaluation workflows
- **Vector / Data: Pinecone**, Weaviate, FAISS, Azure AI Search, PostgreSQL 14.x–16.x, Redis 7.x, Elasticsearch 8.x
- **Cloud / Enterprise: Microsoft Azure**, Azure Functions v4, Azure App Service, Azure DevOps YAML, Docker 24.x, Kubernetes 1.27–1.30, Terraform 1.5–1.8
- **Investment Systems:** Due diligence automation, portfolio analytics, risk monitoring, document intelligence, financial workflow dashboards, stakeholder reporting

EXPERIENCE

Lightedge — Lead Software Engineer — **May2022 – Apr 2026**

Remote/ Full time — *Houston, TX*

- Built enterprise **AI investment workflow applications** with **React 18.x**, **Next.js 13.x–14.x**, **Python 3.10–3.12**, and **FastAPI 0.100.x–0.110.x** for document-heavy due diligence and portfolio intelligence teams.
- Designed production **RAG pipelines** using **Azure OpenAI**, **Azure AI Search**, Pinecone, PostgreSQL 14.x–16.x, and metadata filtering to improve investment document retrieval accuracy by **31%**.
- Implemented agentic research workflows that summarized CIMs, financial statements, portfolio reports, and risk notes while keeping human approval checkpoints for enterprise investment users.
- Resolved a critical hallucination issue by adding source-grounded citations, prompt guardrails, retrieval scoring thresholds, and regression tests that reduced unsupported AI answers by **27%**.
- Created full-stack deal execution dashboards with **Next.js App Router**, server components, typed API clients, and role-based access control for investment analysts and operating partners.
- Migrated older React 17.x modules into **React 18.x** and **Next.js 14.x**, replacing fragile client-side routing with structured layouts and improving page load speed by **34%**.
- Built Python API services with **FastAPI**, Pydantic validation, OpenAPI schemas, async workers, and Redis queues to process large investment documents reliably.
- Fixed production timeout errors in long-running AI workflows by separating ingestion, embedding, retrieval, and generation jobs through background workers and retry-safe queue design.
- Integrated **Azure DevOps YAML**, Docker 24.x, Terraform 1.5–1.8, and Kubernetes 1.27–1.30 to make AI application releases safer across development, staging, and production environments.
- Improved monitoring with **OpenTelemetry**, Azure Monitor, structured logs, and failure dashboards so engineering and product teams could trace latency across LLM, API, and vector layers.
- Collaborated with investment stakeholders to convert manual risk review steps into automated scoring workflows while keeping audit logs, permission boundaries, and compliance-friendly explanations.
- Mentored engineers on **LLM application architecture**, secure API design, frontend performance, and production incident handling across a distributed enterprise delivery model.

Syscom Digital — Staff Software Engineer **Nov 2017 – Apr 2022**
Remote/ Full time *Bucharest, Romania*

- Delivered full-stack financial and enterprise platforms using **React 16.x–17.x**, **Next.js 10.x–12.x**, **Python 3.8–3.10**, Django 3.2–4.0, and FastAPI 0.68.x–0.85.x.
- Built early AI-assisted document classification workflows for finance teams, combining Python services, PostgreSQL 11.x–14.x, Elasticsearch 7.x, and rule-based validation before LLM adoption matured.
- Developed secure portfolio reporting dashboards with React, TypeScript 4.x, REST APIs, JWT authentication, and permission-aware UI states for internal analysts and client-facing users.
- Fixed recurring data mismatch bugs between frontend reports and backend calculations by introducing shared API contracts, schema validation, and automated integration tests.
- Migrated legacy Django 2.x services to **Django 3.2 LTS** and later Django 4.x patterns, improving maintainability and reducing dependency vulnerability exposure by **29%**.
- Implemented workflow automation for approval chains, document review, and task routing, reducing manual analyst coordination work by **22%** across operational teams.
- Optimized slow investment reporting APIs by replacing N+1 queries, adding indexed filters, caching expensive aggregations, and improving average response time by **38%**.
- Integrated Azure Functions v3–v4, Azure Blob Storage, and Azure DevOps pipelines to support scalable document ingestion and scheduled data processing workloads.
- Created reusable frontend component libraries with React, Material UI 4.x–5.x, form validation, and typed API hooks to standardize complex enterprise screens.
- Supported production incidents by tracing API logs, database locks, failed background jobs, and frontend error reports until stable hotfixes and long-term prevention tasks were completed.

Tessenderlo Group — Senior Software Engineer **Aug 2015 – Oct 2017**
Remote/ Full time *Brussel Metropolitan Area*

- Built enterprise operational platforms with **React 15.x–16.0**, AngularJS 1.5–1.6, Node.js 6.x–8.x, Express 4.x, Python 3.5–3.6, and Django 1.11.
- Developed analytics dashboards for commercial, supply chain, and finance users, giving stakeholders faster access to margin, inventory, and operational risk indicators.
- Implemented new approval and reporting features with reusable frontend modules, REST APIs, and SQL-backed workflows that reduced manual spreadsheet preparation by **26%**.
- Resolved production calculation defects caused by inconsistent currency and unit conversions by centralizing business rules inside tested backend services.
- Migrated older AngularJS screens into React 15.x modules where appropriate, allowing the team to modernize high-value workflows without disrupting active users.
- Improved API reliability by adding validation layers, structured error responses, pagination, and database indexes that reduced recurring support tickets by **21%**.
- Created role-based access controls for internal dashboards so finance, operations, and management users could safely access only the workflows assigned to their roles.
- Worked flexibly across frontend, backend, database, and release tasks, helping the team deliver features even when requirements changed late in the sprint.
- Supported production releases by reviewing logs, reproducing user-reported issues, preparing rollback steps, and documenting fixes for future maintenance.

Tessenderlo Group — Software Engineer **Sep 2011 – Jul 2015**
Hybrid/ Full time *Brussel Metropolitan Area*

- Developed internal business applications with JavaScript ES5, AngularJS 1.2–1.4, jQuery 1.8–2.x, Node.js 0.10–0.12, Express 3.x–4.x, and SQL Server 2008 R2.
- Built workflow screens for procurement, finance, and operational reporting, replacing manual Excel-based tracking with web-based forms and searchable records.
- Fixed browser compatibility issues across older enterprise environments by standardizing JavaScript patterns, CSS fallbacks, and server-side validation behavior.
- Implemented reporting APIs and scheduled export jobs that helped business users reduce repeated manual reconciliation work by **24%**.
- Migrated legacy jQuery-heavy pages into more structured AngularJS modules, improving code separation and reducing frontend defect recurrence by **18%**.

- Created reusable validation utilities for forms, date handling, numeric inputs, and permission checks so developers could implement similar workflows faster.
- Improved SQL query performance by adding targeted indexes, simplifying joins, and moving repeated calculations into cleaner service-layer logic.
- Supported deployment troubleshooting by checking IIS settings, application logs, database permissions, and configuration differences between test and production environments.

Tessenderlo Group — Software Developer **Jun 2009 – Aug 2011**
Hybrid/ Full time *Brussel Metropolitan Area*

- Maintained enterprise web applications using JavaScript ES5, jQuery 1.4–1.6, HTML4–HTML5, CSS2–CSS3, SQL Server 2008, and server-rendered application patterns.
- Implemented internal reporting and data-entry improvements that reduced repetitive administrative work by **17%** for business users across operational teams.
- Resolved form submission bugs, session timeout errors, and inconsistent validation messages by improving client-side checks and backend error handling.
- Assisted with migration from older static pages to cleaner modular templates, making future UI changes faster and easier for senior engineers to review.
- Supported QA and release activities by reproducing defects, documenting fixes, validating database changes, and learning production support discipline in enterprise systems.

EDUCATION

KU Leuven — Master’s Degree, Computer Science **Sep 2003 – May 2009**